Earning the Clinician's Trust: An Eight-Question Confidence Framework for Verified Physical AI in Oncology Trials

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10.5281/zenodo.xxxxxxxx

June 15, 2026 (draft-framework, v0.2.0)

Abstract.

[DRAFTING INSTRUCTIONS: write a single paragraph under 250 words, no citations. Process the thesis from `adoption/prompts/prompt-adoption.md` (the eight CONFIDENCE GOALS) and the evidence base carried in the prior review (`review/`: the ten-gate VVUQ suite, 172 of 172 tests). State the three bedside questions and the single claim: a clinician can place calibrated trust in a verified Physical AI system when, and only when, all eight confidence questions are answered, anchored by the Bill (`Kawchak2026HR9510Bill`, DOI 10.5281/zenodo.20619762).]

Disclaimer: This work is independent and not endorsed or sponsored by trial sponsors, FDA, CROs, sites, IRBs, regulators, or medical societies; and was generated using Artificial Intelligence.

Keywords Physical AI | Oncology clinical trials | Verification before generation | VVUQ | Calibrated trust | Automation bias | Clinician adoption | Patient safety

1 Introduction: the questions a clinician asks at the bedside

This framework is written for a reader who treats cancer patients and who will be asked, soon, to supervise an autonomous system in a clinical trial, but who has never driven a surgical robot or written a line of verification code. It defines its terms as it goes, and it organizes the decision to trust around eight plain questions a clinician already asks of any new tool.

A few definitions make the rest readable. *Physical AI* is artificial intelligence that acts in the physical world through a machine, such as a surgical humanoid in an oncology trial, rather than only returning text on a screen. *Verification before generation* is the principle that a machine may propose an action, but the proposal is checked against fixed, published standards before anything reaches a patient. *Claude Code* is the software agent that drafts the candidate action, *Codex* is a second, independent agent that reviews it, and *VVUQ*, meaning verification, validation, and uncertainty quantification, is the ten-gate examination the proposal must pass. The examination resolves in exactly one of three ways: ACCEPT and act under audit, ESCALATE to a qualified human, or BLOCK before execution. *Calibrated trust* is the goal of this framework: reliance proportioned to demonstrated capability, avoiding both *automation bias* (over-trust) and *algorithm aversion* (under-trust).

[DRAFTING INSTRUCTIONS: in the full-framework, expand these definitions with the mechanism described in the prior review (`review/`) and the Bill (`Kawchak2026HR9510Bill`, DOI 10.5281/zenodo.20619762). Anchor calibrated trust, automation bias, and algorithm aversion in `adoption/references/framework_refs.bib` (`lee2004trust`, `goddard2012automation`, `dietvorst2015algorithm`). Reproduce Mermaid figures `adoption/mermaid/diagrams/03-verification-before-generation.md` and `02-eight-confidence-questions.md` as colored `cfwfig` TikZ.]

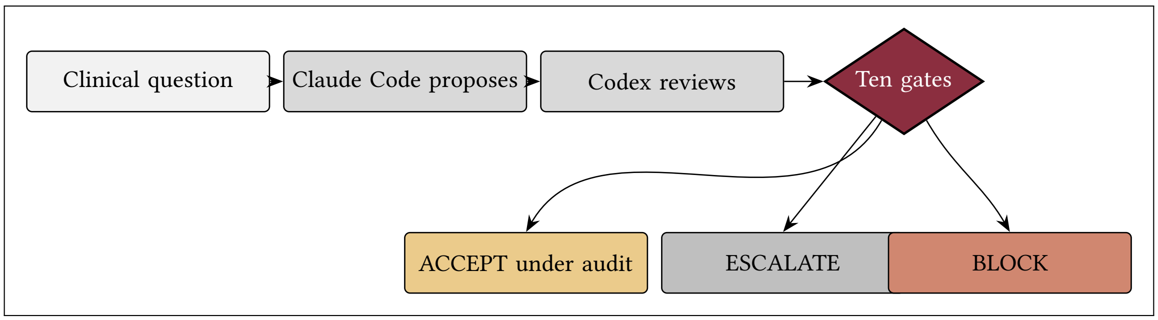

Figure 1. Verification before generation, the clinician's view (placeholder; expanded from Mermaid 03 in the full-framework).

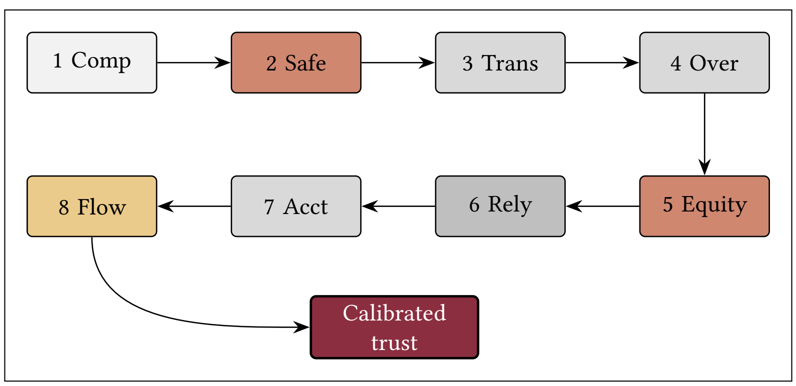

Figure 2. The eight confidence questions (placeholder; expanded from Mermaid 02 in the full-framework). They accumulate into calibrated trust.

Three questions recur throughout, asked politely and answered concretely. Would I let this system near my own family member? Can I defend this decision at tumor board? And if an auditor asked me tomorrow, could I show exactly what happened and why? The remainder takes the eight confidence questions in turn, each defining its terms, pairing its claim with a cited fact, and closing on the specific mechanism that answers it.

Contents

2 Competence: is it actually good at this, or only good on paper?

The first question a clinician asks of any tool is whether it is competent at the specific task, on patients like the ones in this clinic. Competence is not a single benchmark number; it is a stack of evidence built in order, from bench accuracy to prospective validation on the site's own population.

[DRAFTING INSTRUCTIONS: in the full-framework, build the competence case from the author's assurance lineage in `review/references/author_works.bib` (`Kawchak2026MobilePancreaticCancerUnitree`, `Kawchak2026VVUQOncologyClinicalTrial`) and the 172 of 172 automated-test result carried in the prior review. Anchor the human-plus-machine framing in `topol2019high`. Reproduce Mermaid `adoption/mermaid/diagrams/06-competence-evidence-stack.md` and the calibrated-trust band `05-calibrated-trust-quadrant.md` as colored cfwfig TikZ. Carry the ADOPTION ACTION "local validation". Build a body-width table of evidence levels (bench, retrospective, prospective, assurance) with what each establishes.]

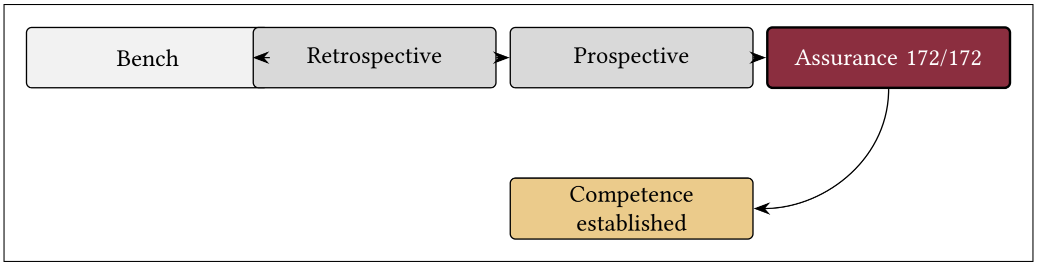

Figure 3. The competence evidence stack (placeholder; expanded from Mermaid 06 in the full-framework).

Competence is answered not by a claim but by evidence the clinician can inspect, and by the discipline of validating on the local population before clinical use.

3 Safety: what stops it before it hurts someone?

The clinician's duty of non-maleficence makes safety the second question. It is not enough that a system usually behaves; the clinician needs to know what concretely stops an irreversible action before it can reach the patient.

[DRAFTING INSTRUCTIONS: in the full-framework, ground the safety case in the two strongest external sources in `adoption/references/framework_refs.bib`: `kohn2000toerr` (To Err Is Human) and `makary2016medical` (medical error as a leading cause of death). Describe the ten gates and the catastrophe-predicate BLOCK carried in the prior review, and the adverse-event response lineage in `review/references/author_works.bib` (`Kawchak2026ThreefoldHumanoid247`). Reproduce Mermaid `adoption/mermaid/diagrams/04-ten` and `07-safety-containment-flow.md` as colored cfwfig TikZ. Carry the KEY TERM "standard operating procedure". Build a body-width table of gate to what it catches.]

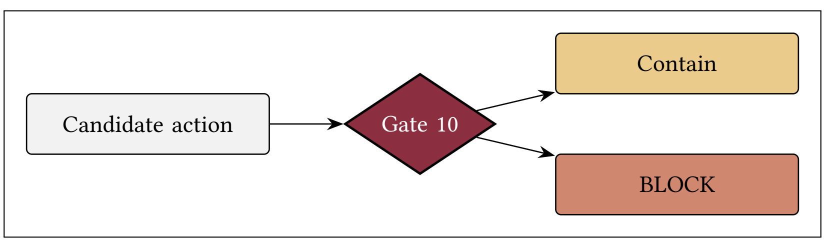

Figure 4. The catastrophe-predicate gate (placeholder; expanded from Mermaid 04 and 07 in the full-framework).

A framework cannot promise that nothing will go wrong. It can promise a hard stop before the worst outcomes, and a record of every action that does proceed.

4 Transparency: can I see why it did that?

A clinician cannot endorse a black box. The third question is whether the system's reasoning is inspectable and whether a complete, tamper-evident record lets the clinician reconstruct events after the fact.

[DRAFTING INSTRUCTIONS: in the full-framework, describe explainable output and the hash-chained audit trail, grounded in the FDA action plan `fda2021samd` and the author's records and server lineage in `review/references/author_works.bib`

(`Kawchak2026NationalMCPServersPhysical`, `Kawchak2026PhysicalAIOncologyClinical`). Reproduce Mermaid `adoption/mermaid/diagrams/08-audit-trail-sequence.md` and `18-tumor-board-defensibility-mindmap.md` as colored cfwfig TikZ. Carry the KEY TERM “audit trail”. Build a body-width table of what the record contains and who can read it.]

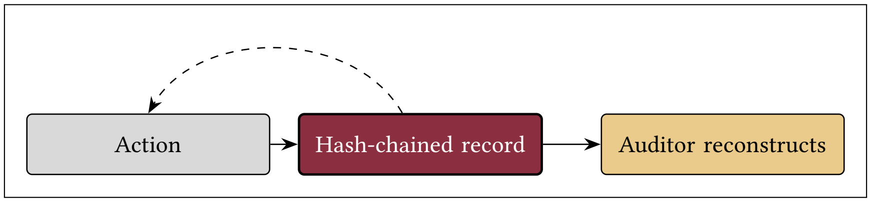

Figure 5. The audit trail (placeholder; expanded from Mermaid 08 in the full-framework).

Transparency is operational, not rhetorical: the test is whether an auditor can reconstruct exactly what happened, and why, from the record alone.

5 Oversight: am I still in control?

The clinician must remain the responsible agent. The fourth question asks whether the human stays in control: how to override, pause, or stop the system, and when it escalates a decision to a qualified person rather than acting alone.

[DRAFTING INSTRUCTIONS: in the full-framework, describe human-in-the-loop and human-on-the-loop supervision, the ESCALATE path, and the unconditional override, grounded in `lee2004trust` (appropriate reliance) and `goddard2012automation` (automation bias in clinical decision support). Reproduce Mermaid `adoption/mermaid/diagrams/09-oversight-loop-state.md`, `14-escalation-pathway-sequence.md`, and `19-override-stop-state.md` as colored cfwfig TikZ. Carry the KEY TERMS “human-in-the-loop” and “human-on-the-loop”. Build a body-width table of oversight modes and the clinician’s controls.]

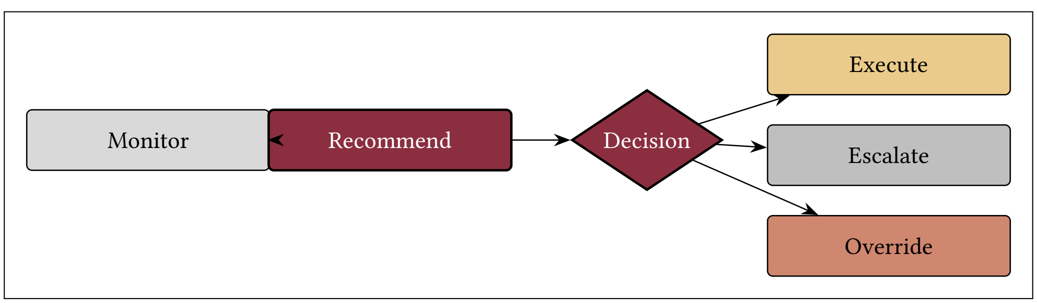

Figure 6. The oversight loop (placeholder; expanded from Mermaid 09 in the full-framework).

Control is not a slogan but a set of switches: a pause, an override, and an escalation path that the clinician can reach at any moment.

6 Equity: does it work for the patient in front of me?

A clinician treats this patient, not the average patient. The fifth question asks whether performance holds for the specific subgroup in the room, and whether bias is actively surveilled rather than assumed away.

[DRAFTING INSTRUCTIONS: in the full-framework, ground the equity case in `obermeyer2019dissecting` (racial bias in a health algorithm) and the subgroup-safeguard framing carried in the prior review. Describe per-subgroup performance reporting and continuous bias surveillance. Reproduce Mermaid `adoption/mermaid/diagrams/10-subgroup-equity-quadrant.md` as colored cfwfig TikZ. Build a body-width table of subgroup audit dimensions (age, ancestry, sex, comorbidity, access) and the metric reported for each.]

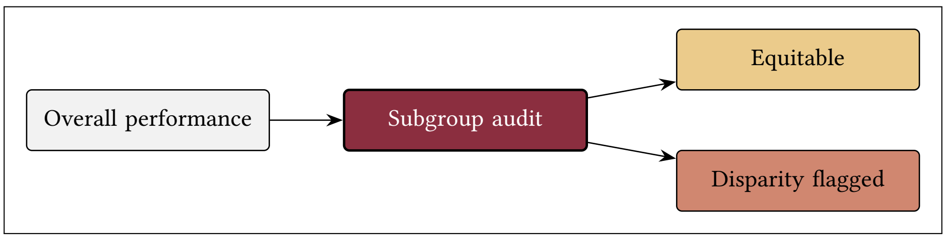

Figure 7. Subgroup equity (placeholder; expanded from Mermaid 10 in the full-framework).

Equity is answered by measurement: performance reported per subgroup, with disparities surfaced as surveillance priorities rather than buried in an average.

7 Reliability: will it be the same tomorrow, at 3 a.m.?

The clinician needs the same answer tomorrow as today. The sixth question asks whether the system behaves consistently over time and load, how model drift is detected and handled, and how uncertainty is quantified and communicated.

[DRAFTING INSTRUCTIONS: in the full-framework, ground reliability in the credibility assessment of `asme2018vv40` (ASME V&V 40) and the uncertainty quantification in the VVUQ suite carried in the prior review. Describe reproducibility controls, drift monitoring and revalidation, and confidence intervals at the point of decision. Reproduce Mermaid `adoption/mermaid/diagrams/15-uncertainty-gate-flow.md`, `16-model-revalidation-gitgraph.md`, and `17-night-shift-sequence.md` as colored cfwfig TikZ. Carry the ADOPTION ACTION "continuous monitoring". Build a body-width table of reliability controls and their triggers.]

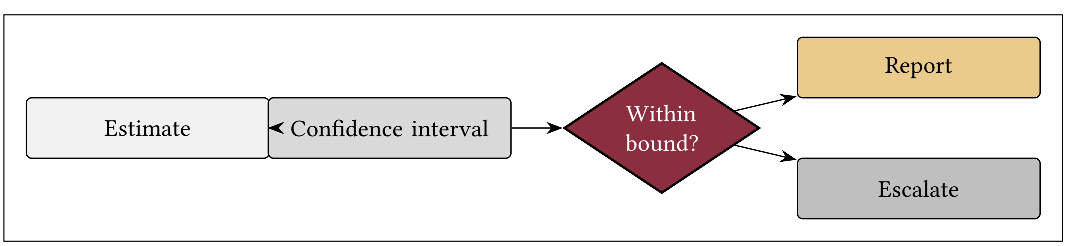

Figure 8. The uncertainty gate (placeholder; expanded from Mermaid 15 in the full-framework).

Reliability is answered by honesty about doubt: a quantified confidence interval, a plan for drift, and a guarantee that 3 a.m. is treated exactly like noon.

8 Accountability: who is responsible when it is wrong?

A clinician must know where responsibility sits. The seventh question asks who answers when something goes wrong, what is documented and by whom, and how liability is allocated across the human and the system.

[DRAFTING INSTRUCTIONS: in the full-framework, describe explicit roles, the signed standard operating procedure, and the documentation chain, grounded in the author's documentation and regulatory-adaption lineage in `review/references/author_works.bib` (`Kawchak2026PhysicalAIOncologyClinical`, `Kawchak2026Adaption21CFRPart`, `Kawchak2026AdaptionICHHarmonisedGuideline`). Reproduce Mermaid `adoption/mermaid/diagrams/13-accountability-raci.md` as colored cfwfig TikZ. Carry the KEY TERMS "standard operating procedure" and "audit trail". Build a body-width RACI table (clinician, sponsor, developer, IRB, auditor) by function.]

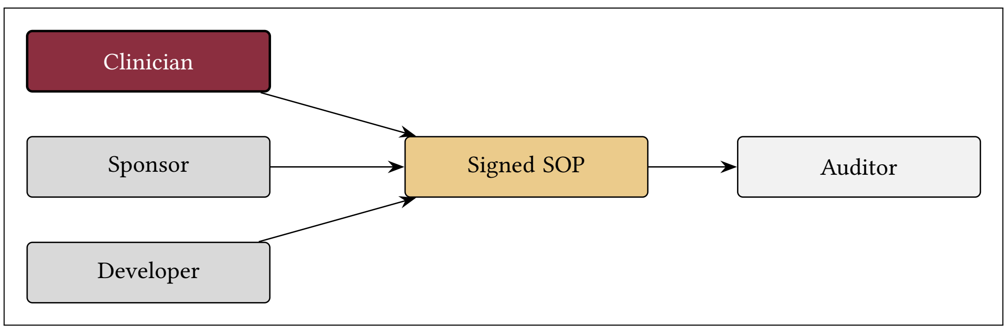

Figure 9. Where accountability sits (placeholder; expanded from Mermaid 13 in the full-framework).

Accountability is answered by drawing it: named roles, a signed procedure, and a documentation chain that an auditor can follow from action to owner.

9 Workflow: does it fit how I actually work?

The best safe tool fails if it does not fit. The eighth question asks whether the system reduces burden or adds to it, and how it changes consent and the clinic day, because a tool that fights the workflow will be worked around or abandoned.

[DRAFTING INSTRUCTIONS: in the full-framework, ground adoption in `dietvorst2015algorithm` (algorithm aversion after seeing error) and `novak2020patient` (patient stories and consent), and in the patient-journey lineage `Kawchak2026CancerPatientsJourney`. Describe workflow integration, burden measurement, and a verified consent process. Reproduce Mermaid `adoption/mermaid/diagrams/11-clinician-workflow-journey.md`, `12-adoption-maturity-timeline.md`, and `20-bedside-to-trial-trust-capstone.md` as colored cfwfig TikZ. Carry the ADOPTION ACTIONS "workflow integration" and "continuous monitoring". Build a body-width table mapping each of the eight questions to the mechanism that answers it (the synthesis table).]

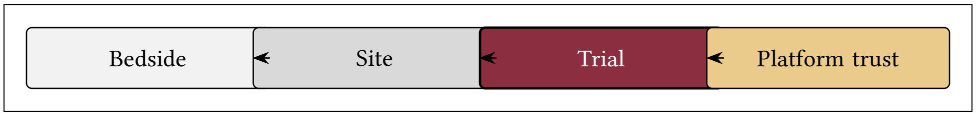

Figure 10. From bedside to trial-wide trust (placeholder; expanded from Mermaid 20 in the full-framework).

Workflow is answered by fit: a system embedded in the clinic day and the consent process, measured for burden, and scaled from one bedside decision to trial-wide calibrated trust.

References

[1] J. D. Lee and K. A. See, "Trust in automation: Designing for appropriate reliance," *Human Factors*, vol. 46, no. 1, pp. 50–80, 2004. https://doi.org/10.1518/hfes.46.1.50_30392.

[2] B. J. Dietvorst, J. P. Simmons, and C. Massey, "Algorithm aversion: People erroneously avoid algorithms after seeing them err," *Journal of Experimental Psychology: General*, vol. 144, no. 1, pp. 114–126, 2015. https://doi.org/10.1037/xge0000033.

[3] K. Goddard, A. Roudsari, and J. C. Wyatt, "Automation bias: a systematic review of frequency, effect mediators, and mitigators," *Journal of the American Medical Informatics Association*, vol. 19, no. 1, pp. 121–127, 2012. https://doi.org/10.1136/amiajnl-2011-000089.

[4] L. T. Kohn, J. M. Corrigan, and M. S. Donaldson, eds., *To Err Is Human: Building a Safer Health System.* Washington, DC: National Academies Press, 2000. Institute of Medicine. https://doi.org/10.17226/9728.

[5] M. A. Makary and M. Daniel, "Medical error - the third leading cause of death in the us," *BMJ*, vol. 353, p. i2139, 2016. https://doi.org/10.1136/bmj.i2139.

[6] Z. Obermeyer, B. Powers, C. Vogeli, and S. Mullainathan, "Dissecting racial bias in an algorithm used to manage the health of populations," *Science*, vol. 366, no. 6464, pp. 447–453, 2019. https://doi.org/10.1126/science.aax2342.

[7] E. J. Topol, "High-performance medicine: the convergence of human and artificial intelligence," *Nature Medicine*, vol. 25, no. 1, pp. 44–56, 2019. https://doi.org/10.1038/s41591-018-0300-7.

[8] U.S. Food and Drug Administration, "Artificial Intelligence/Machine Learning (AI/ML)-Based Software as a Medical Device (SaMD) Action Plan." U.S. Food and Drug Administration, 2021. https://www.fda.gov/medical-devices/software-medical-device-samd/artificial-intelligence-and-machine-learning-software-medical-device.

[9] American Society of Mechanical Engineers, "Assessing Credibility of Computational Modeling Through Verification and Validation: Application to Medical Devices (ASME V&V 40-2018)," 2018. https://www.asme.org/codes-standards/find-codes-standards/v-v-40-assessing-credibility-computational-modeling-verification-validati

[10] A. Ingber, *Regulating Emotion AI in the United States: Insights from Empirical Inquiry.* PhD thesis, Deep Blue Repositories, 2025. Working paper / Doctoral dissertation.

[11] S. Moreland-Russell, C. Barbero, S. Andersen, N. Geary, E. A. Dodson, and R. C. Brownson, ""Hearing from all sides": How legislative testimony influences state level policy-makers in the United States," *International Journal of Health Policy and Management*, vol. 4, no. 2, pp. 91–98, 2015. https://doi.org/10.15171/ijhpm.2015.13.

[12] L. L. Novak, S. George, K. A. Wallston, Y. A. Joosten, T. L. Israel, C. L. Simpson, and C. H. Wilkins, "Patient stories can make a difference in patient-centered research design," *Journal of Patient Experience*, vol. 7, no. 6, pp. 1438–1444, 2020. https://doi.org/10.1177/2374373520958340.

[13] M. Putturaj, ""Honouring the storyteller": The potential of Playback Theatre in health policy and systems research," *Health Policy and Planning*, vol. 40, no. 7, pp. 809–817, 2025. https://doi.org/10.1093/heapol/czad000.

[14] G. Kok, L. K. Bartholomew, G. S. Parcel, N. H. Gottlieb, and M. E. Fernández, "Finding theory- and evidence-based alternatives to fear appeals: Intervention mapping," *International Journal of Psychology*, vol. 49, no. 2, pp. 98–107, 2013. https://doi.org/10.1002/ijop.12001.

[15] K. Kawchak, "H. r. 9510 (bill v5.0) 2026," jun 2026. ORCID record work; source snapshot exported from ORCID webarchive.

[16] K. Kawchak, "H. r. 9510 (bill v4.0) 2026," jun 2026. ORCID record work; source snapshot exported from ORCID webarchive.

[17] K. Kawchak, "H. r. 9510 (bill v3.0) 2026," jun 2026. ORCID record work; source snapshot exported from ORCID webarchive.

[18] K. Kawchak, "Verification before generation act of 2026," jun 2026. ORCID record work; source snapshot exported from ORCID webarchive.

[19] K. Kawchak, "Vvuq physical ai oncology trial bill," may 2026. ORCID record work; source snapshot exported from ORCID webarchive.

[20] K. Kawchak, "Mobile pancreatic cancer unitree h2 surgical humanoid with priority vvuq," may 2026. ORCID record work; source snapshot exported from ORCID webarchive.

[21] K. Kawchak, "Vvuq oncology clinical trial llm verification automation priority over existing generated code," may 2026. ORCID record work; source snapshot exported from ORCID webarchive.

[22] K. Kawchak, "Threefold humanoid 24/7 adverse event oncology trial response team: 4-site rotation," may 2026. ORCID record work; source snapshot exported from ORCID webarchive.

[23] K. Kawchak, "2030: 60 second pancreatic ductal adenocarcinoma robotic whipple procedure & daraxonrasib simulation," may 2026. ORCID record work; source snapshot exported from ORCID webarchive.

[24] K. Kawchak, "2030: 60 second glioblastoma ai robotic surgery," may 2026. ORCID record work; source snapshot exported from ORCID webarchive.

[25] K. Kawchak, "Patient priority of proposed u.s. bills for physical ai oncology clinical trials," may 2026. ORCID record work; source snapshot exported from ORCID webarchive.

[26] K. Kawchak, "Accelerated patient prediction in physical ai oncology clinical trials: 4 extensive simulations," may 2026. ORCID record work; source snapshot exported from ORCID webarchive.

[27] K. Kawchak, "Fully automated sponsor: Physical ai oncology clinical trial platform," apr 2026. ORCID record work; source snapshot exported from ORCID webarchive.

[28] K. Kawchak, "National platform for physical ai oncology trials," mar 2026. ORCID record work; source snapshot exported from ORCID webarchive.

[29] K. Kawchak, "Physical ai oncology clinical trial site complete documentation package," mar 2026. ORCID record work; source snapshot exported from ORCID webarchive.

[30] K. Kawchak, “A cancer patient’s journey through a regulated physical ai oncology trial,” mar 2026. ORCID record work; source snapshot exported from ORCID webarchive.

[31] K. Kawchak, “Adaption: 21 cfr part 312, end-to-end physical ai oncology clinical trial unification,” mar 2026. ORCID record work; source snapshot exported from ORCID webarchive.

[32] K. Kawchak, “Adaption: 21 cfr part 50, end-to-end physical ai oncology clinical trial unification,” mar 2026. ORCID record work; source snapshot exported from ORCID webarchive.

[33] K. Kawchak, “Adaption: Ich harmonised guideline, end-to-end physical ai oncology clinical trial unification,” mar 2026. ORCID record work; source snapshot exported from ORCID webarchive.

[34] K. Kawchak, “National mcp servers for physical ai oncology clinical trial systems,” mar 2026. ORCID record work; source snapshot exported from ORCID webarchive.

[35] K. Kawchak, “Trialmcp: Mcp servers for physical ai oncology clinical trial systems,” mar 2026. ORCID record work; source snapshot exported from ORCID webarchive.

[36] K. Kawchak, “Federated learning physical ai oncology trials unification,” feb 2026. ORCID record work; source snapshot exported from ORCID webarchive.

[37] K. Kawchak, “Unification standard level for physical ai oncology trials,” feb 2026. ORCID record work; source snapshot exported from ORCID webarchive.

[38] K. Kawchak, “Code generation competition: 16 proprietary vs. open-source llms & iterative learning based on fda adverse event reporting system,” dec 2025. ORCID record work; source snapshot exported from ORCID webarchive.

[39] K. Kawchak, “Ai peer review acceleration of llm-generated glioblastoma clinical trial patient matching ml, fda/ich/iso, and fastapi,” dec 2025. ORCID record work; source snapshot exported from ORCID webarchive.

[40] K. Kawchak, “Llm-generated glioblastoma drug synergy machine learning: From rapid code prototypes to project deliverables package,” nov 2025. ORCID record work; source snapshot exported from ORCID webarchive.

[41] K. Kawchak, “End-to-end oncology clinical trial llm efficiency for industry adoption with fda/ich regulations,” oct 2025. ORCID record work; source snapshot exported from ORCID webarchive.

[42] K. Kawchak, “Accelerating fda compliance and cost efficiency of in silico clinical trials via ai digital twin pancreatic cancer simulation,” oct 2025. ORCID record work; source snapshot exported from ORCID webarchive.

[43] K. Kawchak, “Qsp metastatic pancreatic cancer ai clinical trial simulation from protocol to prediction: Code, vvuq, and playbook,” aug 2025. ORCID record work; source snapshot exported from ORCID webarchive.

[44] K. Kawchak, “Chatgpt 100,000 patient 24-month in silico phase iii 5-arm pancreatic cancer clinical trial triplicate,” jul 2025. ORCID record work; source snapshot exported from ORCID webarchive.

[45] K. Kawchak, “End-to-end pancreatic ductal adenocarcinoma digital twin clinical trial proposals,” jun 2025. ORCID record work; source snapshot exported from ORCID webarchive.

[46] K. Kawchak, “10 year glioblastoma clinical trial meta-analyses by autonomous ai at scale. survival, hr, ae, and rob scored in ai reports and charts, including verifications,” may 2025. ORCID record work; source snapshot exported from ORCID webarchive.

[47] K. Kawchak, “Ai revolution toward the cure of lung adenocarcinoma,” apr 2025. ORCID record work; source snapshot exported from ORCID webarchive.

[48] K. Kawchak, “Autonomous llm agent and scalable reasoning llm for generating cancer drug industry cost solutions,” mar 2025. ORCID record work; source snapshot exported from ORCID webarchive.

[49] K. Kawchak, “Cost containment of global monoclonal antibody drugs and cancer clinical trials via llm focused reasoning,” feb 2025. ORCID record work; source snapshot exported from ORCID webarchive.

[50] K. Kawchak, “Clinical decision support based on bevacizumab cancer trials and pushing the limitations of advanced llms,” jan 2025. ORCID record work; source snapshot exported from ORCID webarchive.

[51] K. Kawchak, “Cancer vs. conversational artificial intelligence,” dec 2024. ORCID record work; source snapshot exported from ORCID webarchive.

[52] K. Kawchak, “mab bioprocess engineering in-context table forecasts using conversational ai literature insight generations,” dec 2024. ORCID record work; source snapshot exported from ORCID webarchive.

[53] K. Kawchak, “Monoclonal antibody bioprocess engineering advancements using conversational artificial intelligence,” oct 2024. ORCID record work; source snapshot exported from ORCID webarchive.

[54] K. Kawchak, “Paclitaxel biosynthesis ai breakthrough,” oct 2024. ORCID record work; source snapshot exported from ORCID webarchive.

[55] K. Kawchak, “High dimensional and complex spectrometric data analysis of an organic compound using large multimodal models and chained outputs,” sep 2024. ORCID record work; source snapshot exported from ORCID webarchive.

[56] K. Kawchak, “Lmm spectrometric determination of an organic compound,” aug 2024. ORCID record work; source snapshot exported from ORCID webarchive.

[57] K. Kawchak, “Lmm chemical research with document retrieval,” aug 2024. ORCID record work; source snapshot exported from ORCID webarchive.

Acknowledgments

The author acknowledges Anthropic for access to Claude Code for the instruction, drafting, and assembly of this framework; OpenAI for ChatGPT subject support; and Google Gemini for additional research assistance.

Ethical Disclosures

The author declares no competing interests. This is an independent framework and is not endorsed, sponsored, or approved by any trial sponsor, CRO, site, IRB, regulator, or medical society, nor by CFR, ICH, or FDA. H. R. 9510 is an illustrative bill number; it is not enacted law and this framework is not legal or clinical advice. All supporting numbers are simulation or illustrative results unless tied to a cited source.

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Cite This Framework

Kawchak K. Earning the Clinician's Trust: An Eight-Question Confidence Framework for Verified Physical AI in Oncology Trials. Zenodo. 2026; 10.5281/zenodo.xxxxxxxx.

Data Availability

All sources, figures, and prompt records supporting this framework are openly available in the `kevinkawchak/law-physical-ai-trials` repository under `adoption/` (the prompt, the four sub-prompts, the colored Mermaid catalog, and the three framework stages), and build on the prior review under `review/` and the author's Zenodo deposits. All data sets are synthetic or illustrative; no patient-identifiable information is included.